

Resources

<!DOCTYPE html>

◆ Free resources • Genie Ashwani prioritized

in 6 Months

A focused, project-first roadmap for a beginner targeting Java backend internships and entry-level roles. Learn for 2–3 hours daily, code every day, and publish every project on GitHub.

6

Focused months

2–3h

Daily commitment

3

Portfolio projects

150+

DSA problems

How to follow it

Do not binge-watch. For every hour of learning, spend at least one hour writing, debugging or improving code.

60% Build

25% Learn

01

Java Foundations

Syntax, logic, JVM basics and developer setup

Weeks 1–4



Setup + Java basics

JDK, IntelliJ, JVM/JRE/JDK, variables, data types, operators, input and output.

★ Genie: Java/DSA Lecture 1

★ Genie: JVM Explained

Dev.java Learn



Control flow + methods

if/else, switch, loops, arrays, strings, methods, scope and basic debugging.

★ Genie: Free Java Lectures

Official Java Basics



Problem solving

Dry runs, patterns, number problems, arrays and strings; solve 30 easy questions.

★ Genie DSA Series

HackerRank Java



Git + GitHub basics

Repository, add, commit, push, pull, branches and a clean README.

GitHub Skills

Pro Git

Month-end deliverable: Console-based Student Grade Manager with methods, arrays, validation and a public GitHub repository.

02

Core Java Mastery

OOP, collections, exceptions and modern Java

Weeks 5–8



OOP deeply

Classes, objects, constructors, encapsulation, inheritance, polymorphism, abstraction and interfaces.

★ Genie: OOP videos

Dev.java Classes



Collections framework

List, Set, Map, Queue, generics, Comparable, Comparator, iteration and complexity.

★ Genie: Collections

Java Collections Guide



Exceptions + files

Checked/unchecked exceptions, custom exceptions, try-with-resources and file I/O.

★ Genie: Exceptions

Official Exceptions



Java 8+

Lambdas, functional interfaces, streams, Optional, method references and date/time API.

★ Genie: Java 8

Official Lambdas

Month-end deliverable: Library Management System using OOP, collections, custom exceptions, file persistence and unit-friendly design.

03

Database & Quality

SQL, JDBC, Maven and automated testing

Weeks 9–12



SQL + MySQL

DDL/DML, filters, joins, grouping, subqueries, indexes, normalization and transactions.

★ Genie: SQL search

SQLBolt Interactive



JDBC

Connections, PreparedStatement, ResultSet, CRUD, transactions, DAO pattern and connection safety.

★ Genie: JDBC Full Course

Oracle JDBC Tutorial



Maven + project structure

pom.xml, dependencies, lifecycle, plugins, profiles and standard package structure.

★ Genie: Maven

Maven Getting Started



Testing

JUnit 5, assertions, lifecycle, parameterized tests, Mockito mocks and service-layer testing.

★ Genie: JUnit/Mockito

JUnit 5 Guide

Mockito

Month-end deliverable: Expense Tracker backed by MySQL/JDBC, Maven, DAO pattern and at least 20 meaningful unit tests.

04

Spring Boot APIs

Production-style REST APIs with persistence

Weeks 13–16



Spring Core + Boot

IoC, DI, beans, configuration, auto-configuration, starters, profiles and layered architecture.

★ Genie: Spring Boot One Video

★ Genie: Boot + Project

Spring Guides



REST APIs

Controllers, DTOs, CRUD, HTTP methods/status codes, validation and global exception handling.

★ Genie: REST API

Official REST Guide



Spring Data JPA

Entities, relationships, repositories, pagination, JPQL, transactions, auditing and N+1 awareness.

★ Genie: Spring Data JPA

Official JPA Guide



API tooling

Postman, Swagger/OpenAPI, logging, configuration, pagination and consistent API responses.

★ Genie: Swagger/Postman

Springdoc OpenAPI

Month-end deliverable: Job Portal REST API with users, jobs, applications, validation, pagination, API docs and MySQL.

05

Security & Deployment

Authentication, Docker and microservices basics

Weeks 17–20



Spring Security + JWT

Authentication, authorization, password encoding, filters, roles, JWT access and refresh flow.

★ Genie: Spring Security Playlist

Official Security Docs



Docker

Images, containers, Dockerfile, volumes, networks and Docker Compose for app + database.

★ Genie: Docker

Docker Java Guide



Microservices fundamentals

Service boundaries, REST communication, config, discovery, gateway, resilience and distributed trade-offs.

★ Genie: Microservices Project

Spring Microservices



Deployment basics

Environment variables, Linux commands, build JAR, logs, health checks and one free/cloud deployment.

★ Genie: AWS/Deployment

GitHub Actions

Month-end deliverable: Secure the Job Portal with JWT and roles, containerize it with MySQL, add health checks and deploy a live demo.

06

Job-Ready Sprint

Weeks 21–24

☐ **Capstone project**

Build one polished backend with auth, database, tests, Docker, docs, logging and deployment.

★ Genie: Project videos

Project choices below

☐ **DSA interview set**

Arrays, strings, hashing, linked lists, stacks, queues, recursion, sorting, binary search and trees basics.

★ Genie: Free DSA Batch LeetCode

☐ **LLD + design patterns**

SOLID, Factory, Strategy, Observer, Builder and one design such as Parking Lot.

★ Genie: Design Patterns

★ Genie: Parking Lot LLD

☐ **Resume + interview practice**

One-page resume, GitHub cleanup, project explanation, Java/Spring/SQL questions and mock interviews.

★ Genie: Interview Questions

Java Skill Checklist

Final outcome: 3 pinned repositories, 1 live capstone, 150+ DSA questions, a focused resume and the ability to explain every major technical decision.

Capstone choices

Choose only one. Depth, tests, documentation and deployment matter more than building many incomplete projects.

Recommended

Job Portal API

A practical domain that demonstrates roles, workflows and database relationships.

- Recruiter and candidate roles

Strong portfolio

E-commerce Backend

Shows transactional thinking and separation of business concerns.

- Products, cart, orders and inventory

- Jobs, applications and status workflow
- JWT, search, pagination and email events

- Idempotent checkout simulation
- Admin APIs, validation and tests

Advanced

Course Platform

Excellent for demonstrating access control and modular backend design.

- Courses, lessons and enrollments
- Student/instructor/admin roles
- Progress tracking and reviews

Keep the scope realistic: Kafka, Kubernetes, Redis, Elasticsearch and advanced system design are valuable, but they are not required before the first Java backend role. Add them after the core roadmap or only when your capstone genuinely needs them.

Built as a free learning roadmap. Resource order gives first priority to [Genie Ashwani](#); official Java, Spring, GitHub, SQL and Docker documentation fills topic gaps. Links were checked in July 2026.